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چک پوینت های فصل دوم جاوا



1:- Identify and fix the errors in the following code:

```
1  public class Test {
2    public void main(string[] args) {
3      double i = 50.0;
4      double k = i + 50.0;
5      double j = k + 1;
 6
7      System.out.println("j is " + j + " and  k is " + k);
8        }
9 }
```

در این جا یک کیو رد staticفراموش شده است و sدر Stringحرف کوچک است درست آن قرار ذیل است

```
public class Main {
    public static void main(String[]args) {
       duoble i = 50.0;
       duoble k = i + 50.0;
       duoble j = k + 1;
       System.out.print(" k is " + k + " and j is " + j);
    }
}
```

2:- Which of the following identifiers are valid? Which are Java keywords?

miles, Test, a++, −−a, 4#R, $4, #44, apps class, public, int, x, y, radius
radius , x , y , apps, $4 ,miles- they are valid
class and public are the key word

3:- Identify and fix the errors in the following code:

```
public class Test {
    public static void main(String[] args) {
        int i = k + 2;
        System.out.println(i);
    }
}
```

مشکل کد شماره ۳این است که متغیر kقبلاً تعریف نشده و سیستم او را نمیشناسد

شکل درست آن قرار ذیل است:

```
public class Test {
   public static void main( String[]args) {
     int k = 4;
     int i = k + 2;
     System.out.print(i);
   }
 }
```

4:- Identify and fix the errors in the following code:

```
public class Test {
      public static void main(String[] args) {
          int i = j = k = 2;
          System.out.println(i + " " + j + " " + k);
      }
}
```

مشکل این کد این است که چند علا مه مساوی در یک خط است و متحولین
ذکر شده قبلا تعریف نشده شکل درست آن قرار ذیل است

```
public class Test {
   public static void main(String[]args) {
     int i, j, k;
System.out.print(i + " " + j + " " + k);
   }
}
```

5:- What are the benefits of using constants? Declare an int constant SIZE with value 20.

1) you don’t have to repeatedly type the same value if it is used multiple times;

2) if you have to change the constant value
you need to change it only in a single location in the source code;

3) a descriptive name for a constant makes the program easy to read.

6:- Translate the following algorithm into Java code:

Step 1: Declare a double variable named miles with an initial value 100.

Step 2: Declare a double constant named KILOMETERS_PER_MILE with value 1.609.

Step 3: Declare a double variable named kilometers, multiply miles and KILOMETERS_PER_MILE, and assign the result to kilometers.

Step 4: Display kilometers to the console. What is kilometers after Step 4?

```
public class Main {
  public static void main(String[]args) {
     double mile = 100;
     double KILOMETERS_PER_MILE = 1.609;;
     double kilometers = KILOMETERS_PER_MILE  * 100;
    System.out.print(kilometers);
  }
}
```

answer = 160.9 kilometers

7:- What are the naming conventions for class names, method names, constants, and variables?

Which of the following items can be a constant, a method, a variable, or a class according to the Java naming conventions?

MAX_VALUE, Test, read, readDouble

class start with capital letter, methode start withe lwercase, constant's whole letters are capital, variable start with lowercase.

MAX_VALUE = constant, Test = class, read = methode and v, readDouble = variable and m.

8:- Find the largest and smallest byte, short, int, long, float, and double. Which of these data types requires the least amount of memory?
smallest = byte, largest = long and double

Show the result of the following remainders:

56/6 , 78/-4 , -34/5 ، -34/-5 ، 5/1 , 1/5

```
public class Main {
   public static void main( String[]args) {
      int a = 56%6;
      int b = 78%-4;
      int c = -34%5;
      int d = -34%-5;
      int e = 5%1;
      int f = 1%5;
       System.out.print("R of a" + a + " R of b " + b + " R of c " + c + " R of
       d " + d + " R of e " + e + " R of f " + f);
   }            result = R of a 2 R of b  2 R of c -4 R of d -4 R of e 0 R of f1
}
```

If today is Tuesday, what will be the day in 100 days?

```
public class Main {
    public static void main(String[]args) {
        int a = 100%7;
           System.out.print(a);
     }                  result = 2 ,  if R is = 0 tuesday, if R is = 1
}                       wednesday, if R = 2 thursday
```

day of 100 is tursday

What is the result of 25 / 4? How would you rewrite the expression if you wished the result to be a floating-point number?

we have write it 25.0/4

Are the following statements correct? If so, show the output.

a:- System.out.println("25 / 4 is " + 25 / 4);

b:- System.out.println("25 / 4.0 is " + 25 / 4.0);

result a is not correct and b is correct, reault of a is 6 and result of b is 6.25

9:-How many accurate digits are stored in a float or double type variable?

float = 32bit = 2^32 , double = 64 bit = 2^64

Which of the following are correct literals for floating-point numbers?

12.3, 12.3e+2, 23.4e-2, -334.4, 20.5, 39F, 40D

12.3, 23.4e-2, -334.4, 20.5, they are literals for floating- point number

Which of the following are the same as 52.534?

5.2534e+1, 0.52534e+2, 525.34e-1, 5.2534e+0

5.2534e+1, 0.52534e+2, 525.34e-1 they are same as 52.534

10:- What does REPL stand for? How do you launch JShell?

REPL stand for Read-Evaluate-Print Loop

11:- How would you write the following arithmetic expressions in Java?

$$\frac{4}{3(r+34)} - 9(a + bc) + \frac{3\,d(2 + a)}{a + bd}$$

answer = 4/3(r + 34) - 9*(a + b*c) + 3*d*(2 + a)/a + b*d

12:- How do you obtain the current second, minute, and hour?

```
public class ShowCurrentTime {
   public static void main(String[]args) {
      long totalSeconds = System.currentTime Seconds is ();
      long currentSeconds =totalSecond%60;
      long totalMinutes = totalSeconds/60
      long currentMinutes = totalMinutes%60;
      long totalHours = totalMinutes/60;
      long currentHours = totalHours%60;
       System.out.print("current time is" + current Hours + ":" +
current Minutes + ":" + current Seconds + "GMT");
      }
  }
```

13:- Show the output of the following code:

```
double a = 6.5;
a += a + 1;
System.out.println(a);
```

```
a = 6;
a /= 2;
System.out.println(a);
```

```
public class Main {
  public static void main(String[]args) {
    double a 6.5;
    a +=a + 1;
    System out.println(a);
    a = 6;
    a /=2;
    System.out.println(a);
  }
}
```

result = 14.0
3.0

14:- Show the output of the following code:

```
public class Main {
  public static void main( String[]args) {
    int a = 6;
    int b = a++;
    System.out.println(a);
    System.out.println(b);
    a = 6;
    b = ++a;
    System.out.println(a);
    System.out.println(b);
  }
}
```

result = 7
6
7
7

15:- How would you write the following arithmetic expression?

$$\frac{-b+\sqrt{b^2 - 4ac}}{2a}$$

answer = (-b+√b^2-4*a*c)/2*a

16:- How do you write a statement to let the user enter a double value from the keyboard?

answer= double radiuse = input.nextDouble();

اگر قبل ازین بیانیه inputرا به عنوان Scannerساخته باشیم می توانیم توسط این بیانیه اعداد اعشاری را از یوزر بگیریم.

- What happens if you entered 5a when executing the following code?

جواب: برنامه نمی‌تواند آن را به یک مقدار doubleتبدیل کند؛ چون 5aیک عدد معتبر نیست.

17:- Write a statement to display the result of 2^3.5.

```
public class Main {
  public static void main(Strung[]args) {
    System.out.println(Math.pow(2, 3.5));
  }
}
```

how do you write mr^2? System.out.print(m*Math.pow(r, 2));

18:- Which of the following are correct literals? 5_2534e+1, _2534, 5_2, 5_

5_2534e+1,5_2 are true and _2534 , 5_ are false

Which of these statements are true?

a. Any expression can be used as a statement.
b. The expression x++ can be used as a statement.
c. The statement x = x + 5 is also an expression.
d. The statement x = y = x = 0 is illegal.

answer = a false
b true
c true
d false

۱۹:- فورمول محمد بن موسی را در جاوا چگونه نوشته می کنید

جواب System.out.print(b + Math.sqrt(b*b -4*a*c)/(2*a))

نکته: در این جا به جای جذر از sqrtاستفاده میکنیم.

20:- Can you declare a variable as int and later redeclare it as double?

this is not allowed

```
int x = 2;
double x = 6;
```

Will overflow cause a runtime error? (no)

در intجاوا معمولاً عدد به شکل دیگری در محدوده عددی برمی‌گردد، بدون اینکه برنامه متوقف شود.

٢١:- Can different types of numeric values be used together in a computation?

answer = yes

Java allows different numeric types, such as int and double, to be used in the same computation.

example:

```
public class Main {
    public static void main(String[]args) {
        double x =12.8;
        int y = (int)x;
        System.out.print(y);
    }
}
```

output = 12

22:- If you change (int)(tax * 100) / 100.0 to (int)(tax * 100) / 100 in line 11 in Listing 2.8, what will be the output for a purchase amount of 197.556?

The output will be: Sales tax is $11

٢٣:- Show the output of the following code:

```
float f = 2.5F;
int i = (int)f;
System.out.println("f is " + f);
System.out.println("i is " + i);
```

```
public class Main {
   public static void main(String[]args) {
      float f = 2.5F;
      int i = (int)f;
      System.out.println("f is " + f);
      System.out.println("i is " + i);
   }
}
```

out put: f is 2.5 and i is 2

٢٤:- Show the output of the following code:

```
double amount = 5;
System.out.println(amount / 2);
System.out.println(5 / 2);
```

```
public class Main {
    public static void main(String[]args) {
        double amount = 5;
        System.out.println(amount / 2);
        System.out.println(5 / 2);
    }
}
```

out put: 2.5
2

Write an expression that rounds up a double value in variable d to an integer.

```
public class Main {
    public static void main(String[]args) {
        double d = 4.3;
        int result = (int)Math.ceil(d);
        System.out.println(result);
    }
}
```

out put = 5

نکته: در این pdf سوالات چک پوینت به ترتیب
نیست چون من اول سوالات ساده را حل کردم
و بعد سوالات مشکل را در آخر حل کردم